

# Joel Thomas

## Full Stack Developer

Košice, Slovakia | +421 907 943 418 | 8281o29196@gmail.com  
GitHub: [github.com/joelt006](https://github.com/joelt006) | Portfolio: [joelt006.github.io/Portfolio-site](https://joelt006.github.io/Portfolio-site)

## SUMMARY

Final-year Informatics student and full-stack web developer who has built and deployed complete applications end to end, from a feature-rich digital banking platform to a cloud-hosted app on Azure, plus a shipped feature in an active 3D visualization framework as part of a bachelor thesis. Strong across the stack: Python / Django REST back ends, Vue and React front ends, relational databases, authentication and security, and containerized cloud deployment. Eager to bring this hands-on engineering experience to a professional development team.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, SQL, HTML, CSS, GLSL (shaders)

**Back End:** Django, Django REST Framework, Flask, RESTful APIs, JWT / OTP auth, WebSockets, PostgreSQL, SQLite

**Front End:** Vue 3 (Vuex, Vue Router), React, Three.js, A-Frame, WebXR, Axios, responsive CSS

**Cloud & DevOps:** Docker, Microsoft Azure (ACI, ACR, Storage), Caddy reverse proxy, HTTPS / Let's Encrypt, shell scripting

**Practices:** Git & GitHub, secure auth design, atomic DB transactions, technical documentation, debugging, testing

**Currently Learning:** TypeScript, PHP, Node.js

## EDUCATION

### B.Sc. in Informatics

Expected June 2026

*Technical University of Košice, Slovakia*

- Final-year informatics student; coursework in software development, data structures, algorithms, and databases.
- Bachelor thesis: "Recent Approaches to Data Visualisation in Virtual Reality" — designed and implemented a 3D statistical-error visualization feature for an immersive, browser-based particle-physics framework.

## PROJECTS

### MyBank — Full-Stack Digital Banking Platform

*Django REST Framework, Vue 3, Vuex, SimpleJWT, SQLite*

- Built a complete banking application covering accounts, money transfers, loans, fixed deposits, card management, notifications, and a customer support system, with separate customer and admin portals.
- Implemented layered security: JWT authentication, email OTP two-factor login, CAPTCHA, bcrypt-hashed transaction PINs, rate limiting, and active-session management.
- Designed atomic money transfers using database row locking (`select_for_update`) to guarantee balance integrity, plus scheduled transfers, beneficiaries, and CSV statement export.
- Created a real-time notification system and live financial calculators (loan EMI, FD maturity) for an interactive front-end experience.

### CloudNotes — Cloud-Deployed Note-Taking App

*Flask, PostgreSQL, Docker, Microsoft Azure (ACI / ACR), Caddy*

- Deployed a containerized web app to Azure Container Instances running three coordinated containers (reverse proxy, Flask API, PostgreSQL) in a single group.
- Automated the full deployment with idempotent shell scripts that provision Azure resources, build and push Docker images, and configure persistent storage and backups.
- Configured automatic HTTPS via Caddy and Let's Encrypt, container auto-restart, and graceful database-connection retry on startup.

### 3D Error-Interval Visualization for NDMVR-Core

Bachelor Thesis

*JavaScript, Three.js, A-Frame, WebXR, JSROOT, RxJS, GLSL*

- Designed and shipped a feature for an active immersive web framework that visualizes statistical uncertainty on 3D histograms, using instanced GPU geometry and custom shaders to render thousands of bins per draw call at VR frame rates; integrated with its reactive (RxJS) architecture and fully documented.